

DiaTool-DPO

Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

A novel DPO-based method that models TA-LLM interactions as an MDP with 5 dialogue states and 3 query types, automatically constructing preference pairs to improve slot-filling and tool call rejection.

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Core Problem

TA-LLMs trained only with SFT struggle with:

- Incomplete queries (missing arguments)
- Out-of-scope requests
- Hallucinated tool invocations

Three Failure Modes

1. No Follow-up Questions

Model fails to ask clarifying questions when arguments are missing

2. Hallucinated Tool Calls

Model invokes tools with fabricated arguments instead of requesting info

3. Failed Tool Rejection

Model accepts out-of-scope requests instead of politely declining

Need for Preference Optimization:

SFT with expert trajectories alone is insufficient to learn conversational skills like slot-filling and appropriate tool call rejection. DiaTool-DPO uses Direct Preference Optimization to enhance dialogue capabilities by automatically constructing paired trajectory datasets of correct and incorrect dialogue flows.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



Key Properties:

- States are internal (unobserved) — the model learns to navigate them implicitly via preference pairs
- State transitions are conditioned on both user query content and query type (Type 1/2/3)
- This MDP formulation enables systematic, automatic construction of preference pair datasets

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Condition	Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	Sufficient info	"Show me flights from NYC to LA tomorrow morning" — all required arguments provided
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Incomplete info	"Book a flight" — missing origin, destination, date; model asks N follow-up questions
Type 3	$1 \rightarrow 1$	Out-of-scope	"Call my mom using the flight booking tool" — function unavailable; model rejects politely

Why This Matters:

Each query type requires a different dialogue strategy. Type 1 demands efficient direct execution, Type 2 requires adaptive slot-filling questioning, and Type 3 demands appropriate rejection. The MDP formulation allows DiaTool-DPO to construct preference pairs that teach the model to distinguish and correctly respond to each trajectory type — without human annotation.

Automatic Paired Trajectory Construction Without Human Labels

Core Idea: Pair chosen (correct) trajectories with rejected (incorrect) ones. Each pair teaches a specific lesson about dialogue control.

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count & Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089 Easy
Type 1	1→3→4→5	1→(2N)→3→4→5	Prevent redundant slot-filling	562 Hard
Type 1	1→3→4→5	1→(2M)→3→4→5	Prevent redundant slot-filling	2,530 Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090/562 Easy/Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089 Easy
Type 2	1→(2N)→3→4→5	1→3→4→5	Prevent slot hallucination	562 Hard
Type 2	1→(2N)→3→4→5	1→(2M)→3→4→5	Prevent slot hallucination	2,530 Hard
Type 2	—	1→1	Tool call accept	2,089/562 Easy/Hard
Type 3	1→1	1→3→4	Tool call reject	567 Hard
Type 3	1→1	1→(2N)→3→4	Tool call reject	562 Hard

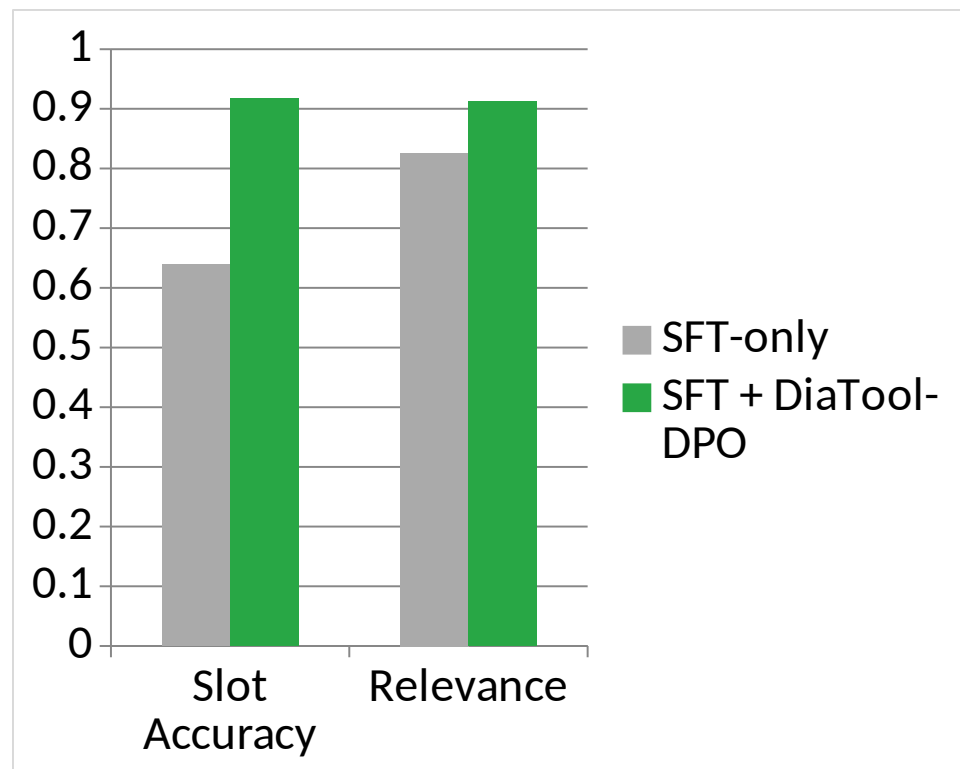
SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Macro Avg.
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.904

Key Finding: SFT + DiaTool-DPO consistently achieves the highest Macro Avg across all three model sizes.

- Prop.-8B: 0.856 → 0.868 (+1.4%) | Prop.-3.1B: 0.791 → 0.818 (+3.4%) | LLaMA-3-8B: 0.816 → 0.904 (+10.8%)
- DPO alone is insufficient — DiaTool-DPO-only performs poorly, confirming DPO complements SFT, not replaces it.

LLaMA-3-8B: +43.5% Slot Accuracy & +10.5% Relevance with DiaTool-DPO



Performance Comparison



Impact: With DiaTool-DPO, LLaMA-3-8B achieves near GPT-4o-level conversational tool use (94.8% info gathering, 91% tool rejection), demonstrating that RL-based dialogue control can unlock GPT-4o-level performance on much smaller open-source models.

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs